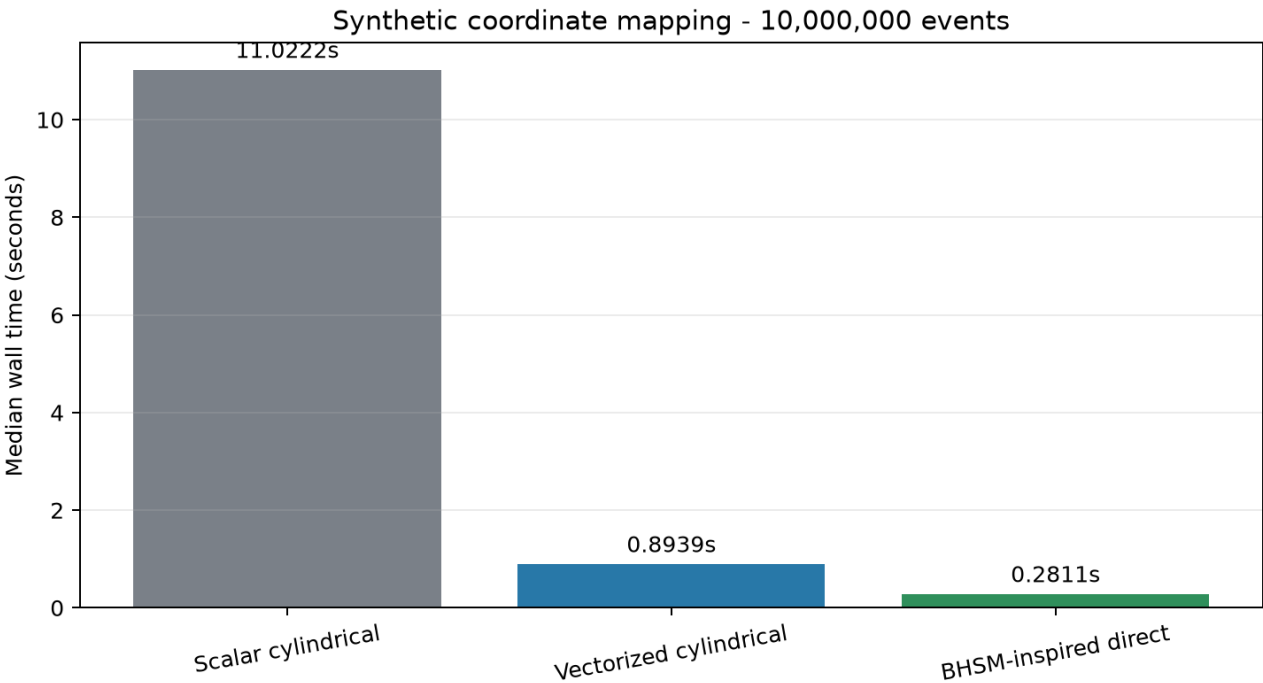


BHSM Synthetic Coordinate-Method Benchmark

Controlled Python/NumPy microbenchmark. This is not production HEP tracking validation.

10,000,000 events; 35% boundary/edge cases; 3 isolated repeats.

Kernel	Median (s)	Events/s	Peak RSS (MiB)
Scalar cylindrical	11.022200	907,260	1159.0
Vectorized cylindrical	0.893868	11,187,328	1931.5
BHSM-inspired direct	0.281142	35,569,214	1617.0



Microbenchmark only; not production HEP tracking validation.

Scalar/BHSM-inspired speedup: **39.205x**. Vectorized-control/BHSM-inspired speedup: **3.179x**.

Peak RSS reduction: **-39.5%** versus scalar and **16.3%** versus vectorized control. Maximum numerical delta: **1.066e-14**.

Branch counters: `HARDWARE_COUNTERS_UNAVAILABLE_ON_WINDOWS`. No branch-misprediction value is inferred.